

$j_1 = 1$	$j_2 = \frac{1}{2}$	$J = \frac{3}{2}$	$J = \frac{1}{2}$
		$M = \frac{3}{2} \quad \frac{1}{2} \quad -\frac{1}{2} \quad -\frac{3}{2}$	$\frac{1}{2} \quad -\frac{1}{2}$
$m_1 = 1$	$m_2 = \frac{1}{2}$	1	
	$-\frac{1}{2}$	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{2}{3}}$
$m_1 = 0$	$m_2 = \frac{1}{2}$	$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$
	$-\frac{1}{2}$	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{1}{3}}$
$m_1 = -1$	$m_2 = \frac{1}{2}$	$\sqrt{\frac{1}{3}}$	$-\sqrt{\frac{2}{3}}$
	$-\frac{1}{2}$	1	

Pion	Nucleon	$I = \frac{3}{2}$	$I = \frac{1}{2}$
		$I_3 = \frac{3}{2} \quad \frac{1}{2} \quad -\frac{1}{2} \quad -\frac{3}{2}$	$\frac{1}{2} \quad -\frac{1}{2}$
π^+	p	1	
π^+	n	$\sqrt{\frac{1}{3}}$	$\sqrt{\frac{2}{3}}$
π^0	p	$\sqrt{\frac{2}{3}}$	$-\sqrt{\frac{1}{3}}$
π^0	n	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{1}{3}}$
π^-	p	$\sqrt{\frac{1}{3}}$	$-\sqrt{\frac{2}{3}}$
π^-	n	1	

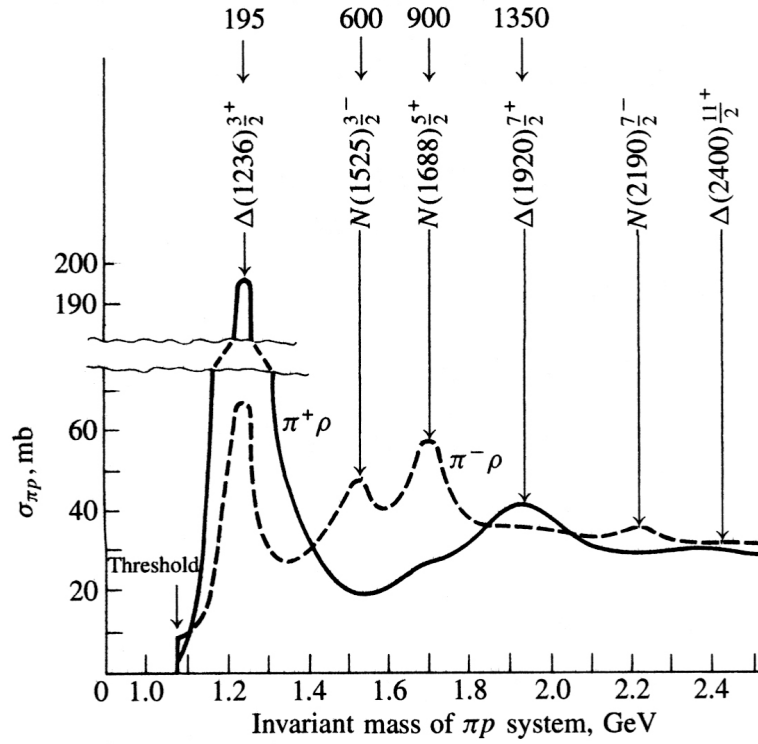


Fig. 3.8. The total cross-section for π^+ and π^- mesons on protons, as a function of pion energy or invariant mass of the pion-nucleon system. The symbol Δ refers to $I = \frac{3}{2}$ resonances, while N refers to $I = \frac{1}{2}$ states. The positions of a few of the known states, together with spin-parity assignments, are given. The four numbers at the top give the pion kinetic energy in MeV.